Hardware User Manual

EtherCAT Series

PCIe-Rxx05-ECAT

AJINEXTEK AXT

Product Information

Full information about other AJINEXTEK products
is available by visiting our Web Site at:
Home Page : www.ajinextek.com
E-mail : support@ajinextek.com

Useful Contact Information

Customer Support Seoul
Tel : 82-31-436-2180~2Fax: 82-31-436-2183

Customer Support Daegu
Tel : 82-53-593-3700~2 Fax: 82-53-593-3703

AJINEXTEK's sales team is always available to assist you in making your decision the final choice of boards or systems is solely and wholly the responsibility of the buyer. AJINEXTEK's entire liability in respect of the board or systems is as set out in AJINEXTEK's standard terms and conditions of sale

ⓒ Copyright 2018 AJINEXTEK co.ltd. All rights reserved.

Contents

1. Overview _____ 오류! 책갈피가 정의되어 있지 않습니다.

1.1. Exordium ..... 오류! 책갈피가 정의되어 있지 않습니다.
1.2. Apply ..... 오류! 책갈피가 정의되어 있지 않습니다.
1.3. Function & Special feature **6**

2. Product Specifications _____ 오류! 책갈피가 정의되어 있지 않습니다.

2.1. Product shape ..... 오류! 책갈피가 정의되어 있지 않습니다.
2.2. Electrical and Physical Specifications ..... 오류! 책갈피가 정의되어 있지 않습니다.

3. Product installation _____ 오류! 책갈피가 정의되어 있지 않습니다.

3.1. Hardware Installation ..... 오류! 책갈피가 정의되어 있지 않습니다.
3.2. Software Installation ..... 오류! 책갈피가 정의되어 있지 않습니다.

4. Product configuration information 오류! 책갈피가 정의되어 있지 않습니다.

4.1. Product connection diagram ..... 오류! 책갈피가 정의되어 있지 않습니다.
4.2. Slave module information ..... 오류! 책갈피가 정의되어 있지 않습니다.
4.3. Cable information ..... 오류! 책갈피가 정의되어 있지 않습니다.

5. Term Description _____ 오류! 책갈피가 정의되어 있지 않습니다.

Revision History

Manual	PCB	Library	Comments
Rev. 1.0, issue 1.0	Rev v1.0	EzSoftware UC v2.0.6.5000	Jul. 27, 2018

	This symbol is intended to alert you to caution (including warnings). If you ignore this warning and act, it means that the board may be damaged due to a defect or defect.

	It contains notes and information on using the board.

1. Overview

1.1. Introduction

This product is a motion control master board based on EtherCAT communication protocol. Up to 32 axes motion and various IO modules can be connected up to 32.

The PCIe-Rxx05-ECAT board communicates in real time with distributed control elements using the EtherCAT communication protocol.

The application field of products enables precise control of step motors, DC servo motors and AC servo motors used as main actuators in industrial equipment such as semiconductor manufacturing equipment, textile equipment, packaging equipment, and PCB engraving machine. The EtherCAT Slave products supported by our company include motion function modules (N3ECAT-PM2Q, N3ECAT-PM4Q), digital input / output modules (N3ECAT-DI32, N3ECAT-DO32T, N3ECAT-DB32T) , N3ECAT-AO8).

1.2. Apply

The application areas of the products are as follows

- ▶ Semiconductor manufacturing equipment
- ▶ Motion Vision
- ▶ Robotics Engineering
- ▶ X-Y table
- ▶ Winding machine
- ▶ Embroidery machine
- ▶ X-Y position control
- ▶ Loading / Unloading
- ▶ Step motor control
- ▶ Machine control
- ▶ Roll Feeding
- ▶ Pick & Placing
- ▶ Research & Development

1.3. function & special feature

▶ **EtherCAT Slave module connection**

- Up to 64 modules can be connected (Motion 32 axis, 32 nodes of IO module)
- Step / Servo Motion, Digital I / O, Analog I / O, Pulse out

▶ **Motion control axis number and IO connection module number**

- PCIe-R0805-ECAT: Motion x 8axis + IO Module x 32 Node
- PCIe-R1605-ECAT: Motion x 16 axis + IO Module x 32 Node
- PCIe-R3205-ECAT: Motion x 32 axis + IO Module x 32 Node

▶ **Advanced motion control function**

- Supports linear interpolation (supports up to 8 axes), circular interpolation
- 3D Helical, Spline support
- Gantry, ECAM, EGEAR support
- Blending motion support
- Support vibration suppression function
- 1D, 2D position correction table support

▶ **Flexible motion product registration**

- CiA-402 Device Profile Support Step / Servo Motion product supports product registration quickly

▶ **Communication transmission cycle**

- Supports 0.5ms to 4msec (Default 1msec)

▶ **Communication jitter**

- 1usec or less (based on DC synchronization)

▶ **Various topology support**

- Line, Star, Ring configuration support

▶ **Support real time motion profile monitoring**

- Real-time graph check of various motion drive profiles and IO output operation status with Motion Scope function Support PCI Express Slot Type
- Support for installation on systems with PCI Express X1 Lane Bus

▶ **Various software support**

- Provides EzSoftware UC program installation package to support equipment development from various operating environment configuration.

2. Specifications

2.1. Product shape



Image 1. Product Photos

2.2. Electrical and Physical Specifications

Table 1. PCIe-Rxx05-ECAT Master Board Specifications

Item	Specification
Supported Communication Protocols	EtherCAT
Connect	100Mbps, STP cable(above CAT.6)
Transmission frequency	0.5msec ~ 4msec(Default 1msec)
Maximum control axis number	32Axes
LED display	Com(Yellow), Error(Red), Power(Yellow)
BUS connection method	PCI Express x 1, Gen 1
Operating temperature range	Temperature: 0 ~ 50℃
Operating humidity range	Humidity: 5 ~ 95% (Non-condensation)
Size	182 (W) mm x 127 (L) mm
Front Panel Connector	RJ45 * 2EA
Operating power supply voltage	12V
Maximum current consumption	Max 800mA

3. Product Installation

3.1. Hardware Installation

After confirming the components of the product, install the board in the following order.

① Before touching the product, discharge the static electricity accumulated in the body.

② Turn off the computer and unplug the power cord.

③ Select the PCI Express slot where the board will be mounted.

④ If the bracket is assembled on the front panel of the computer, remove the bracket.

⑤ Attach the board to the PCI express slot and screw it on the top of the bracket.

⑥ Make sure that the product is installed in close contact with the naked eye as shown in the following figure.

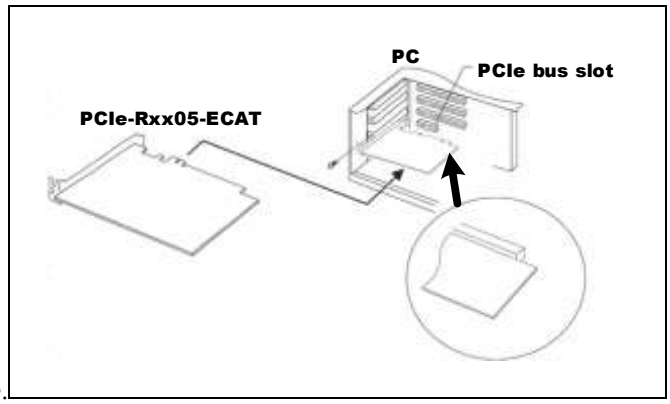


Image 2. Mounting and Checking the Product Slot

① Connect the power plug of the computer and turn on the power switch to operate the system.

② Check if the computer is booting normally. If it does not boot normally, unplug and re-install the product.

⚠	- The ambient temperature inside the computer must be kept within 0 ~ 50 ℃. - It is not recommended to use fanless system. - Depending on the PCI express slot specification of the computer, the board may not be recognized normally, and a general system for industrial use is used.

3.2. Software Installation

To use the product, you need to install EzSoftware UC software.
The software can be downloaded from the separate CD or homepage.

	- Please visit the website http://www.ajinextek.com Download the program.

After installing the EzSoftware UC software, check the board operation in the following procedure.

① Check the normal registration of the device device driver of the installed product in the Device Manager.

② Run "EzSoftware UC" shortcut icon on the desktop.

③ The EzConfig program will be executed and check if the same product is installed.

④ For more information on how to use the program, refer to the program manual.

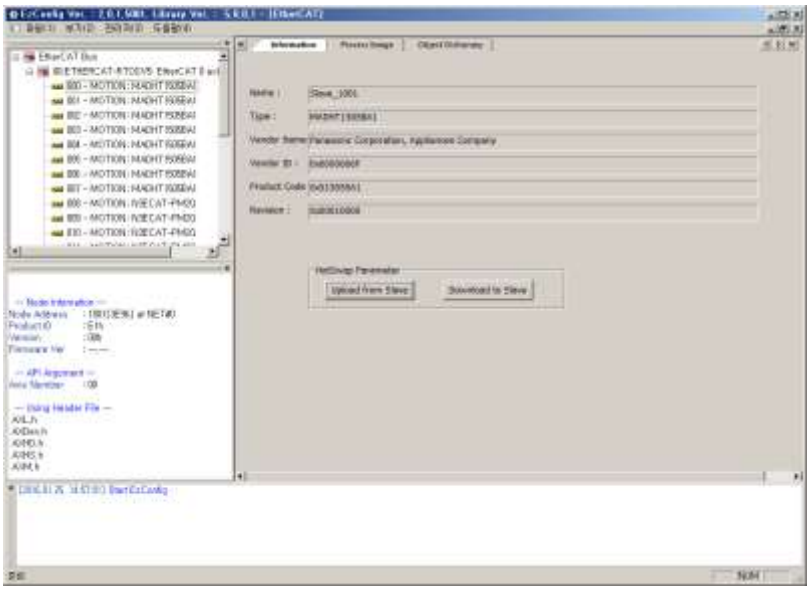


Image 3. EzConfig Program execution screen

4. Product Configuration Information

4.1. Product connection diagram

Connect the product to the peripheral EtherCAT Slot Module as shown in the following figure.

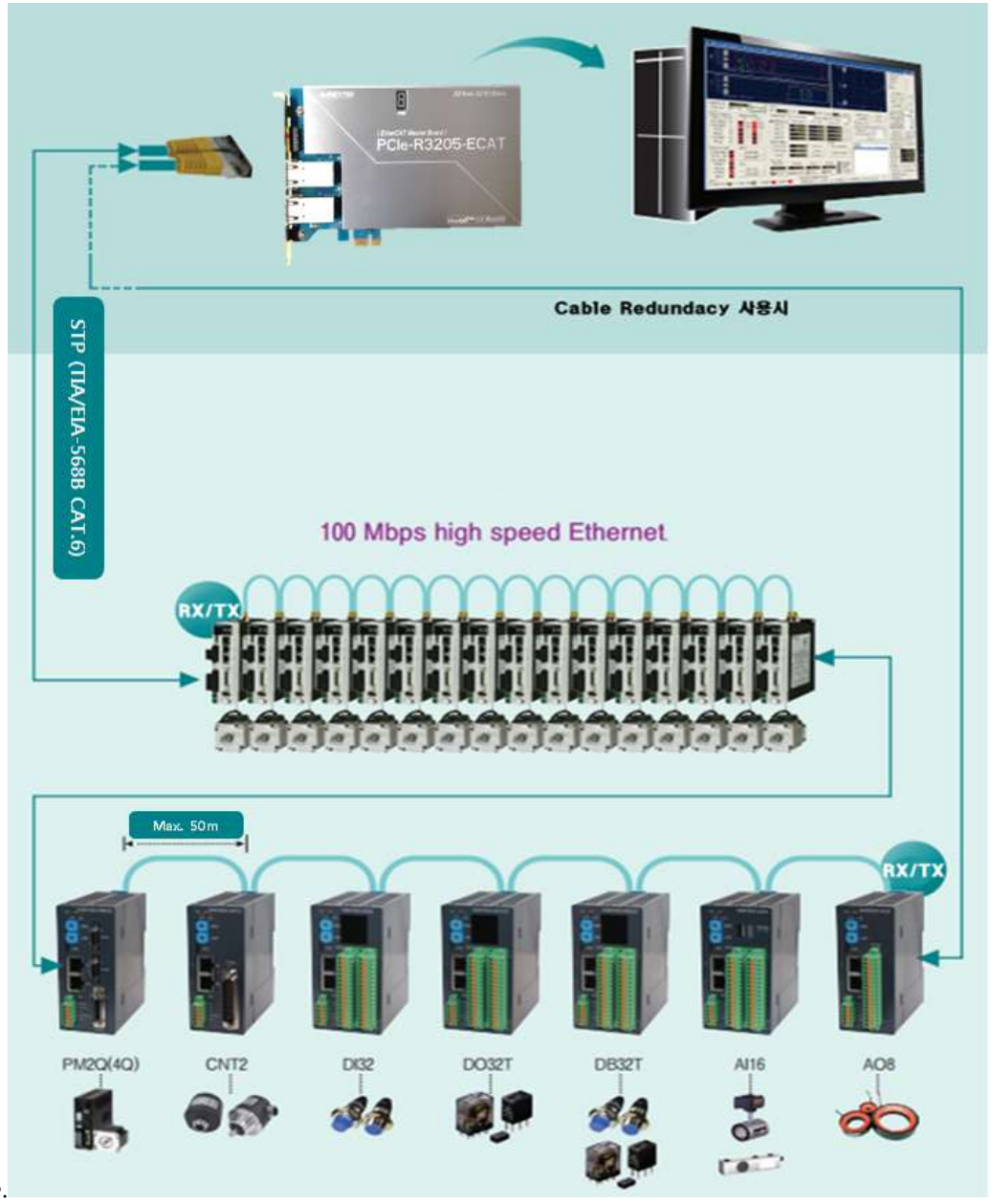


Image 4. Product Structure

4.2. Slave module information

product name	function	form
N3ECAT-PM2Q	Motion 2 Axis Step/Servo driver connect EtherCAT Not supported Servo connect	
N3ECAT-PM4Q	Motion 4 Axis Step/Servo driver connect EtherCAT Not supported Servo connect	
N3ECAT-DI32	Digital Input 32 point 24V NPN(OC) output	
N3ECAT-DO32T	Digital Output 32 point 24V Sink input	
N3ECAT-DB32T	Digital Input 16 point 24V NPN(OC) output & Digital Output 16 point 24V Sink input	
N3ECAT-AI16	Analog Input 16 Channel ±10V, 16-bit resolution. 500uSec Sampling time.	
N3ECAT-AO8	Analog Output 8 Channel ±10V, 16-bit resolution. 500uSec output rate	

Reference : EtherCAT Refer to the corresponding product manual for a detailed description of the slave module.

4.3. Cable information

The network communication cable must use a reliable CAT.6 STP (shielded twisted pair) cable.

In general network communication cable, various external noises generated in the vicinity of communication cable, such as strong wire, relay, actuator, etc., may enter into the communication cable and cause product malfunction.

The maximum length of the communication cable between master and slave or between slave and slave nodes is 50 meters.

The following is the pinmap information of the network communication cable. Cable color is based on EIA / TIA T568B standard.

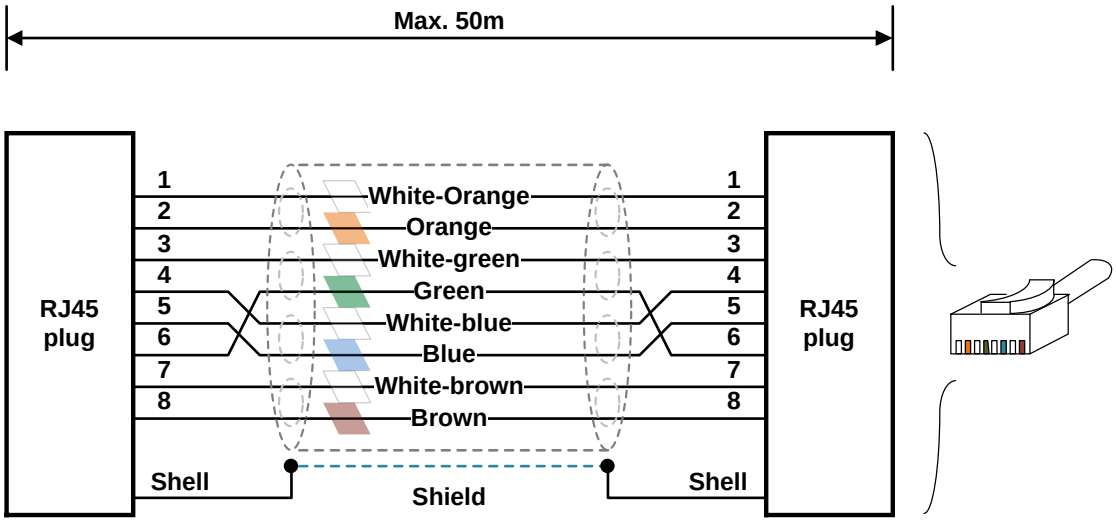


Image 5. Communication cable specification

⚠	- The network communication cable must use the cable that meets the rating. - Do not wire the network communication cable together with a strong wire or actuator drive cable (minimum spacing of at least 30 cm is recommended) - The maximum length of the network communication cable may vary depending on the specifications of the EtherCAT slave module, so be sure to read the corresponding product manual before use.

5. Glossary of Terms

AnyMotion :

This is a motion control module that controls various step motors and servo motors supplied by Ajinomex.

AnyDIO :

It is a digital input / output control module that controls various digital input / output sensors supplied by Ajinexec.

AnyAIO :

It is analog input / output control module which controls various analog input / output sensors supplied by AJINEXTECH.

AnyCOM :

It is a communication control module that controls various kinds of communication supplied from Ajinomex.

AXL :

This is an integrated library used to control various devices supplied by Ajinexec.

EzConfig, EzMotion, EzDIO, EzAI, EzAO, EzCOM Agent :

It is a program that can easily control various devices supplied by Ajinxec which is configured in the system. The EzConfig program provides a function to check connected devices on the board and to search slave modules connected to the network. The EzMotion program is used for motion control. The EzDIO, EzAI, and EzAO programs are used for digital input / output and analog input / output control. And EzCOM program is used for communication control.

The contents of this manual are subject to change without notice. The companies, organizations, products, people and events used in the examples are not actual data. No association with any real company, agency, product, person, or event is intended or should be inferred. It is the responsibility of the user to comply with the applicable copyright laws. No part of this manual may be reproduced or transmitted in any form or by any means (electronic, mechanical, photocopying, or otherwise) without the express written permission of Ajinomoto Co., May not be copied, stored, or introduced into or transmitted to a search system.

Ajinextek Co., Ltd. may have patents, trademarks, copyrights or other intellectual property rights related to this manual. In addition to the rights expressly provided to you by AJINEXTEK under the written license agreement, the furnishing of this manual does not allow you to use any of these patents, copyrights or other intellectual property rights.